Mikołaj Jędrzejewski

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Education

Warsaw University of Technology Warsaw, PL
B.S. in Computer Science and Information Systems (GPA: 4.31 / 5.0) *Oct 2022 – Jun 2026 (Expected)*

- **Core Coursework:** Operating Systems, Algorithms & Data Structures, Numerical Methods, Artificial Intelligence, Databases, Discrete Mathematics, Digital Systems.

National University of Singapore (NUS) Singapore
Exchange Student, Computer Science *Aug 2024 – May 2025*

- **Advanced Electives:** Foundations of Modern Cryptography (CS4230), Blockchain and Distributed Ledger Technologies (IS4302), Parallel and Concurrent Programming (CS3211), Theory of Computation (CS3231), Programming Language Concepts (CS2104), Computer Graphics (CS3241).

Technical Skills

Languages: C, C++ *(Advanced)*; Python, Java, C#, Go, TypeScript, SQL, Circom *(Intermediate)*; Rust *(Familiar)*
Tools & Devops: Git *(Advanced)*; Hardhat, Foundry, Make, LaTeX *(Intermediate)*; Docker, Kubernetes, CMake *(Familiar)*
Frameworks: Spring Boot, Next.js, React, Django

Projects

Decentralized Voting System (Bachelor's Thesis) | *Solidity, Circom, TypeScript* May 2025 – Feb 2026

- Engineered a universally verifiable voting dApp using **ZK-SNARKs (Groth16)** and **exponential ElGamal encryption** on the Baby JubJub curve to ensure ballot secrecy and anonymity.
- Implemented **ERC-4337 Account Abstraction** using **Alchemy** and a custom Paymaster to abstract gas fees, enabling a frictionless "Web2-like" user experience.
- Designed modular circuits in **Circom** and optimized constraint counts for efficient **client-side proving** (WASM), ensuring user privacy without trusted servers.
- Deployed to **Sepolia** and **Base Sepolia (L2)**, achieving a 99% reduction in gas costs vs Ethereum L1.
- Integrated an **NFT module** that mints verifiable election result certificates.

Concurrent Programming Portfolio | *C, C++, Go, Rust* Jan 2025 – May 2025

- Architected a high-throughput trading system in **Go** utilizing a **fan-out pattern**: dedicated goroutines per instrument and channel-based messaging to eliminate lock contention.
- Designed an Order Matching Engine in **C++** using a **Trie-based** order book structure and fine-grained mutex locking for thread-safe order execution.
- Engineered a custom reliable transport protocol in **C** over UDP, implementing a **Sliding Window** mechanism to ensure end-to-end reliability with low overhead.
- Developed an asynchronous TCP server in **Rust** using the **Tokio** runtime to handle 1000+ concurrent client connections via non-blocking I/O.

Web Frontend for SQLancer | *Spring Boot, Next.js, PostgreSQL* Jan 2025 – May 2025

- Extended **SQLancer** by architecting a full-stack Bug Tracking System in a 5-person agile team.
- Engineered a **Spring Boot** backend to orchestrate SQLancer instances, automatically capturing detected logic bugs and persisting them to **PostgreSQL**.
- Developed a **Next.js** dashboard featuring a "GitHub-style" issue workflow, allowing developers to label, inspect, and mark database inconsistencies as resolved.

Carbon Credit DEX | *Solidity, Hardhat, TypeScript* Jan 2025 – May 2025

- Developed contracts utilizing **OpenZeppelin** to mint **ERC-20** carbon credits and **ERC-721** project NFTs.
- Engineered a fixed-rate Liquidity Provider contract to facilitate automated trading and ensure price stability.
- Wrote comprehensive unit tests and traffic simulation scripts using **Hardhat** and **ethers.js**.

Extracurricular & Awards

Optiver READY TRADER GO Challenge | *Top 32 Teams Global* 2023

Polish Olympiad in Informatics | *2nd Stage* 2019, 2020

PW Data Science Club Oct 2022 – Mar 2023

- Collaborated on Exploratory Data Analysis (EDA) and model optimization for Kaggle competitions.